14

Spinal Orthoses

CRISTINA M. SANDERS, MERRIT R. CRUMP, AND JOHN W. NORBURY

SUMMARY PEARLS

- The primary goals of modern spinal orthoses are to restrict mobility, support the spine to aid a weakened muscle group, allow for healing, reduce pain, and/or correct or prevent progression of spinal deformity.
- The process of prescription, fabrication, and fitting of spinal orthoses can be complicated, so a well-developed plan of care and clear goals are necessary.
- A firm understanding of spinal anatomy and spinal biomechanics is necessary when prescribing spinal orthoses.
- Emerging technologies are helping to maximize the efficiency of design and use of spinal orthoses.

History of Spinal Orthotic Management

Orthosis is Greek for "to make straight,"[3] and the first evidence of the use of spinal orthoses can be traced back to Galen (c. AD 131-201) when primitive orthotic devices were made of readily available items (Fig. 14.1), including leather, whalebone, and tree bark. Ambroise Paré (1510–1590) wrote about bracing and spinal supports. Nicolas Andry (1658–1742) coined the term *orthopaedia,* predecessor to the field of orthotics,[2] pertaining to the straightening of children.[6] He focused specifically on how unstable areas, such as fractures, were often held in a corrected position with an orthosis for healing to occur. The primary goal of modern orthoses is to aid a weakened muscle group or correct a deformed body part. The orthoses, including spinal orthoses, can protect body parts to prevent further injury or can correct the position of the body parts in the immediate term or long term. The clinician's priority is to determine which of the spinal motions to control. Good clinical outcomes can be maximized through the proper selection, use, and application of the orthosis.

Terminology

Terminology is often misused in the field of orthotics, and the definitions of some terms commonly used in the field include:

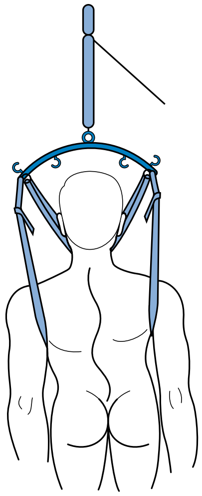
- *Orthosis:* A singular device used to aid or align a weakened body part
- *Orthoses:* Two or more devices used to aid or align a weakened body part(s)
- *Orthotics:* The field of study of orthoses and their management
- *Orthotic:* An adjective used to describe a device (e.g., an orthotic knee immobilizer); this term is improperly used as a noun (e.g., "the patient was fitted with a foot orthotic")
- *Orthotist:* A person trained in the proper fit and fabrication of orthoses
- *Certified orthotist (e.g., American Board for Certification in Prosthetics and Orthotics):* For a person to become a certified orthotist, extensive training in the proper fit and fabrication of orthoses is required. After the education and residency are completed, a national examination for certification can be taken in the United States, supplied by the American Board for Certification in Prosthetics and Orthotics.

Acronyms are frequently used to describe orthoses. They are named for the parts of the body where they are located and have some influence on the motion in that body region. Some examples of spinal orthoses include:

- *CO:* Cervical orthosis
- *CTO:* Cervicothoracic orthosis
- *CTLSO:* Cervicothoracolumbosacral orthosis
- *TLSO:* Thoracolumbosacral orthosis
- *LSO:* Lumbosacral orthosis
- *SO:* Sacral orthosis

Prefabricated or Custom Orthoses

The availability of prefabricated orthoses today presents the rehabilitation team with a variety of choices and some challenges. Many prefabricated orthoses come in various sizes and can be fitted to patients often with little or no adjustment. Although this can be a benefit to the patient and the team in terms of time, care



• **Fig. 14.1** Traction device, 1889.

should be taken to ensure that the design and function of these orthoses are appropriate for the patient's condition and not used purely for convenience. Custom orthoses typically provide a more comfortable fit with a higher degree of control, and they can be designed to accommodate a patient's unique body shape or deformities. Recognition of the time needed to fabricate the orthosis, the experience of the fabricator, the patient's specific condition, and the expectations of the patient are all factors that should be considered when ordering a custom orthosis.

Orthotic Prescription

The orthotic prescription allows for improved communication between clinicians, and it can serve as a justification for funding of the orthosis. Insurer-requested justification for the orthosis is becoming more common as medical costs have increased. Insurer approval of the prescription is more likely if the orthosis clearly increases independence or helps prevent detrimental outcomes, such as falls. Although the prescribing rehabilitation physician is responsible for the final order, open lines of communication and mutual respect between the orthotist and the clinician are crucial for a successful orthotic prescription. The prescription should be accurate and descriptive but not so descriptive as to limit the orthotic team's independent ability to maximize functionality and patient acceptance.

Prescriptions should include the following items: patient identifiers, date, date the orthosis is needed, diagnosis, functional goal, orthosis description, and precautions. Prescriptions should include a justification, such as the correction of alignment, to decrease pain, or to improve function. Brand names and eponyms for the orthosis should be avoided. Established acronyms are acceptable (e.g., TLSO). Detailed descriptions of the orthosis, the joints involved, and the functional goals are important. Before the prescription is finalized, input from the patient, physician, therapist, and orthotist is needed. It is especially important for physicians to review the use, or lack of use, of past orthoses because this will help guide their new prescription. If a patient discontinues the use of an orthosis prematurely, the reason why this occurred should be investigated by the provider before additional resources are expended. Knowledge of the patient's medical condition(s) is essential for a number of reasons. For example, the condition might be progressive, with further expected functional loss. By contrast, the condition might be expected to improve partially or completely in the future.

Spinal Anatomy

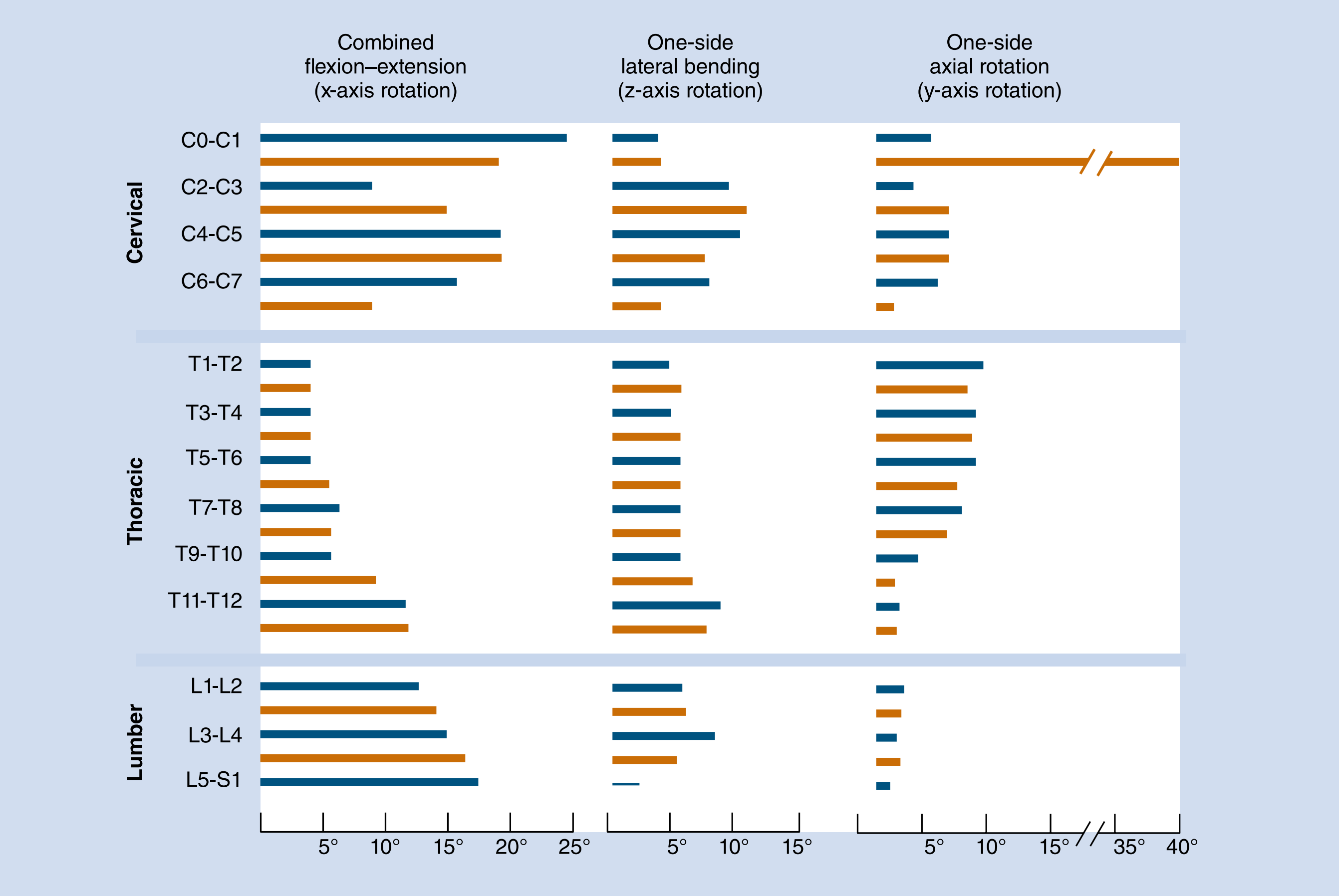
The vertebral column is composed of 33 vertebrae, including 7 cervical, 12 thoracic, 5 lumbar, 5 inferiorly fused vertebrae that form the sacrum, and 5 coccygeal. The spinal column not only bears the weight of the body, but it also allows motion between body parts and serves to protect the spinal cord from injury. Before birth, there is a single C-shaped concave curve anteriorly. At birth, infants have only a small angle at the lumbosacral junction. As a child learns to stand and walk, lordotic curves develop in the cervical and lumbar regions (age 2 years). These changes can be attributed to the increase in weight bearing and differences in the depth of the anterior and posterior regions of the vertebrae and discs.[33,38]

The cervical vertebrae are small and quadrangular, except for C1 and C2 that have some unique features. The cervical articular processes face upward and backward, or downward and forward. The orientation of the facet joints is important to note because it relates to limitations of movement of the vertebral column. The thoracic vertebrae have heart-shaped bodies. The thoracic vertebrae are intermittent in size but increase in size caudally. This is related to the increased weight-bearing requirements. The dorsal length of the thoracic vertebrae is approximately 2 mm more than the ventral side, which could account for the thoracic curve. Their superior articular processes face backward and outward, and the inferior ones face forward and inward. The lumbar vertebrae have a large, kidney-shaped body. Their upper articular processes face medially and slightly posteriorly, and the lower ones face laterally and slightly anteriorly. The five sacral vertebrae are fused in a solid mass and do not contain intervertebral discs. The sacral bony structure acts as a keystone, and weight bearing increases the forces that maintain the sacrum as an integral part of the spinal pelvic complex.

The intervertebral disc is composed of a nucleus pulposus, annulus fibrosus, and cartilaginous end plate. Discs make up approximately one-third of the entire height of the vertebral column. The nucleus contains a matrix of collagen fibers, mucoprotein, and mucopolysaccharides. They have hydrophilic properties, with a very high water content (90%) that decreases with age.[32] The nucleus is centrally located in the cervical thoracic spine, but more posteriorly located in the lumbar spine. The annulus fibrosus has bands of fibrous laminated tissue in concentric directions, and the vertebral end plate is composed of hyaline cartilage.[15]

Normal Spine Biomechanics

Movement of the vertebral column occurs as a combination of small movements between vertebrae. The mobility occurs between the cartilaginous joints at the vertebral bodies and between the articular facets on the vertebral arches. Range of motion is determined by muscle location, tendon insertion, ligamentous limitations, and bony prominences. In the cervical region, axial rotation occurs at the specialized atlantoaxial joint. At the lower cervical levels, flexion, extension, and lateral flexion occur freely. In these areas, however, the articular processes, which face anteriorly or posteriorly, limit rotation. In the thoracic region, movement in all planes is possible, although to a lesser degree. In the lumbar region, flexion, extension, and lateral flexion occur, but rotation is limited because of the inwardly facing articular facets.[18] An understanding



• **Fig. 14.2** Representative values for range of motion of the cervical, thoracic, and lumbar spine as summarized from the literature. (Data from White AA, Panjabi MM. The lumbar spine. In: White AA, Panjabi MM, eds. *Clinical Biomechanics of the Spine.* 2nd ed. Lippincott; 1990.)

of the three-column concept of spine stability/instability is helpful to ensure that the proper orthosis is prescribed. The anterior column consists of the anterior longitudinal ligament, anterior two-thirds of the annulus fibrosus, and the anterior half of the vertebral body. The middle column consists of the posterior longitudinal ligament, posterior one-third of the annulus fibrosus, and the posterior half of the vertebral body. The posterior column consists of the interspinous and supraspinous ligaments, the facet joints, lamina, pedicles, and spinous processes. When the middle column and either the anterior or posterior column are compromised, the spine may be unstable.[10,11]

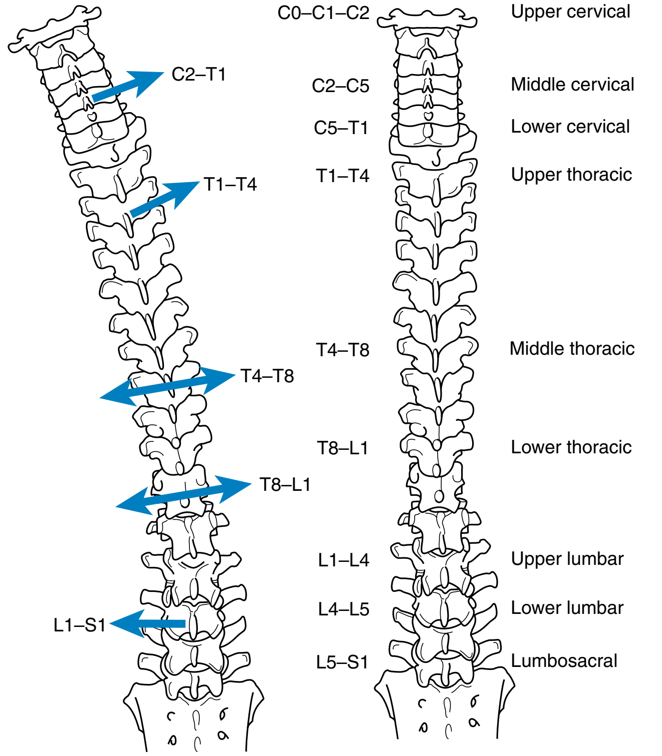
Spine motion can be classified with reference to the horizontal, frontal, and sagittal planes. Spinal motion can shift the center of gravity, which is normally located approximately 2 to 3 cm anterior to the S1 vertebral body. White and Panjabi[41] have provided a summary of the current literature, revealing motion in flexion and extension, laterally, and axially (Fig. 14.2). In the cervical spine, extension occurs predominantly at the occipital C1 junction. Lateral bending occurs mainly at the C3 to C4 and C4 to C5 levels. Axial rotation occurs mostly at the C1 to C2 levels. In the thoracic spine, flexion and extension occur primarily at the T11 to T12 and T12 to L1 levels. Lateral bending is fairly evenly distributed throughout the thoracic levels. Axial rotation occurs mostly at the T1 to T2 level, with a gradual decrease toward the lumbar spine. The thoracic spine is the least mobile because of the restrictive nature of the rib cage. In the lumbar spinal segment, movement in the sagittal plane occurs more at the distal segment, with lateral bending predominantly at the L3 to L4 level. There is insignificant axial rotation in the lumbar spinal segment.

Knowledge of the normal spinal range of motion helps in understanding how the various cervical orthoses can limit that range (Table 14.1). Soft collars provide very little restriction in any plane. The Philadelphia-type collar mostly limits flexion and extension. The four-poster brace has better restriction, especially with flexion-extension and rotation. The halo brace and Minerva body jacket have the most restriction in all planes of motion. An interesting phenomenon related to movement in the spine occurs during motion. If the movement along one axis is consistently associated with movement around another axis, coupling is occurring. For example, if a patient performs left lateral movement (frontal plane) motion, the middle and lower cervical and upper thoracic spine rotate to the left in the axial plane (Fig. 14.3). This causes the spinous processes (posterior side of the body) to move to the right. In the lower thoracic spinal segment, left lateral movement in the frontal plane can cause rotation in the axial plane, with the spinal processes moving in either direction. The lumbar area has a contradictory movement pattern when compared with the cervical spine. With left lateral bending of the lumbar spine, the spinous processes move to the left. A three-dimensional (3D) perspective is important to maintain during examination.

TABLE 14.1 Normal Cervical Motion From Occiput to First Thoracic Vertebra and the Effects of Cervical Orthoses

	MEAN OF NORMAL MOTION (%)		
Orthosis	Flexion or Extension	Lateral Bending	Rotation
Normal[a]	100.0	100.0	100.0
Soft collar[a]	74.2	92.3	82.6
Philadelphia collar	28.9	66.4	43.7
Sternal occipital mandibular immobilizer orthosis	27.7	65.6	33.6
Four-poster brace	20.6	45.9	27.1
Yale cervicothoracic brace	12.8	50.5	18.2
Halo device[a]	4.0	4.0	1.0
Halo device[b]	11.7	8.4	2.4
Minerva body jacket[c]	14.0	15.5	0

[a]Data from Johnson RM, Hart DL, Simmons EF, et al. Cervical orthoses: a study comparing their effectiveness in restricting cervical motion in normal subjects. *J Bone Joint Surg Am.* 1977;59:332.
[b]Data from Lysell E. Motion in the cervical spine, thesis. *Acta Orthop Scand.* 1969;(123):S1.
[c]Data from Maiman D, Millington P, Novak S, et al. The effects of the thermoplastic Minerva body jacket on cervical spine motion. *Neurosurgery.* 1989;25:363-368.



• **Fig. 14.3** Regional coupling patterns. Summary of the coupling of lateral bending and axial rotation in various subdivisions of the spine. In the middle and lower cervical spine, as well as the upper thoracic spine, the same coupling pattern exists. In the middle and lower thoracic spine, the axial rotation, which is coupled with lateral bending, can be in either direction. In the lumbar spine, the spinous processes go to the left with left lateral bending. (Redrawn from White AA, Panjabi MM. The lumbar spine. In: White AA, Panjabi MM, eds. *Clinical Biomechanics of the Spine.* 2nd ed. Lippincott; 1990.)

Patients with scoliosis and patients who undergo radiological testing would benefit from an evaluation for the normal coupling patterns noted.

Differences in Adult and Pediatric Spinal Orthoses

Pediatric and adult orthoses both aim to exert forces on the spine, but there are key differences of indications and goals. Adult spinal columns are less compliant and consequently less responsive to spinal alignment correction. For this reason, adult orthoses are often used to temporarily improve spinal alignment, whereas pediatric bracing is often used to regulate spinal growth before adulthood. For example, corrective orthoses are often utilized in children with scoliosis or kyphosis. It is also more common for orthoses to be used for pain relief in adults compared with the pediatric population.[12]

Description of Orthoses

Head Cervicothoracic Orthoses

Halo Orthosis

Biomechanics. The halo orthosis (Fig. 14.4) provides flexion, extension, and rotational control of the cervical region. Pressure systems are used for control of motion and to provide slight distraction for immobilization of the cervical spine.

Design And Fabrication. The halo orthosis consists of prefabricated components, such as a halo ring, pins, uprights

• **Fig. 14.4** Halo orthosis.

(or superstructure), and vest. The halo ring is fixed to the outer table of the skull bones with generally four or more metal pins. On the typical adult patient, the pins are optimally placed with the patient under local anesthesia, less than 1 inch above the lateral third of each eyebrow (to avoid sinuses) and less than 1 inch above and just posterior to the top of each ear. Upright bars or superstructure connects the ring to a rigid plastic thoracic vest, which is lined with lamb's wool.

The halo is adjustable for flexion, extension, anterior and posterior translation, rotation, and distraction. The vest wraps around the thoracic region of the spine and is fastened laterally, usually by buckles. The design is used to effectively immobilize the cervical spine.

This orthosis provides maximum restriction in motion of all the cervical orthoses. It is the most stable orthosis, especially in the superior cervical spine segment. A halo is used for approximately 3 months (10–12 weeks) to ensure healing of a fracture or of a spinal fusion and is ultimately removed once follow-up x-rays confirm cervical fracture stability per the surgeon's assessment. Usually, a cervical collar is indicated after the halo is removed because the muscles and ligaments supporting the head become weak after disuse. All pins on the halo ring should be checked to ensure tightness 24 to 48 hours after application, and they should be retorqued if necessary.

Indications. The halo is generally used for unstable cervical fractures or postoperative management.

Contraindications and Complications. This orthosis is not indicated for stable fractures or when less invasive management could be used. Patients with an extremely soft skull might not tolerate the pin placement. Furthermore, one must be aware of intersegment snaking that involves flexion of one vertebral segment with extension of the adjacent vertebral body. One must also assess for any signs of clicking, grating, or creaking noises, which may indicate loosening of the Halo pins. A sensation of a "tight fit" halo may also indicate lose pin sites. Other symptoms of pin site loosening include headaches and pin site pain. Daily pin site care is also crucial with normal saline or a topical antiseptic such as povidone-iodine to reduce risk for infection. Infectious complications include pin site cellulitis and brain abscesses, which may present with visual disturbances, fevers, headaches, and seizures.

Special Considerations. Skull density determines halo pin placement as well as the number of halo pins to be used. Although four pins are used on average, more can be necessary in soft skulls (e.g., osteoporotic, fractured, or in an infant) to distribute the force over a broader area of the skull. The use of halo devices in older patients has become more controversial because the halo orthosis has been associated with respiratory compromise, aspiration, and an 8% mortality in this population.[36]

A 2005 study compared halo vest immobilization in the elderly compared with younger patients; they concluded that halo vest immobilization in patients older than 65 years was associated with the worst outcomes, irrespective of injury severity. Younger populations with age less than 45 years tend to do well with halo vest immobilization. Another retrospective study in 2009 compared halo vest immobilization in a population of 409 cervical spine injury patients over an 8-year period at a Level 1 trauma center. Despite high complication rates with pin site infections (39%) and instability (38%), treatment with halo vest immobilization was successful in up to 85% of patients, proving that halo vest immobilization is a reasonable treatment option for most unstable cervical spinal injuries.[5,26]

Cervical Orthoses

Type: Philadelphia, Miami J, and Aspen Collar

Biomechanics. The Philadelphia (Fig. 14.5), Miami J (Fig. 14.6), and Aspen cervical orthoses provide some control of flexion, extension, and lateral bending but minimal rotational control of the cervical region. Pressure systems are used for control of motion and to provide slight distraction for immobilization of the cervical spine. Circumferential pressure is also intended to provide warmth and serve as a kinesthetic reminder for the patient.

Design and Fabrication. These orthoses are prefabricated, consisting of one or two pieces that are usually attached with Velcro straps. Two-piece designs have an anterior and posterior section. The anterior section supports the mandible and rests on

• **Fig. 14.5** Philadelphia orthosis.

• **Fig. 14.6** Miami J orthosis.

the superior edge of the sternum. The posterior aspect of the collar supports the head at the occipital level.

Indications. They are used primarily for cervical sprains, strains, or stable fractures. They are also frequently worn after anterior cervical fusions and anterior discectomy procedures and are less frequently used for Jefferson fracture and midcervical traumatic spondylolisthesis. They can also be used for protection and for limited mobility after surgery to allow healing. Miami J collars are composed of a polyethylene material, allowing for decreased skin temperature, thus reducing the chance of skin breakdown and diaphoresis. Some clinicians utilize Miami J collars for patients dealing with tracheostomy care secondary to respiratory insufficiency, while others prefer the Aspen collar because of its large anterior opening.

It is important to note that cervical collars are primarily worn for stabilization of the spine, and do not improve patient outcomes over time. In fact, prolonged wearing of cervical collars could prolong return to work even in benign whiplash injuries.[28]

Contraindications. In cadaver models, these orthoses have been found to be insufficient for immobilizing the unstable spine.[19] The cervical orthoses tend to lose effectiveness at higher and lower cervical levels (occiput–C2 and C6–C7), and thus a more restrictive device may be appropriate when there are concerns at those upper and lower cervical levels.[1]

Type: Soft Cervical Collar

This orthosis is often used as a kinesthetic reminder for patients to limit their neck motion. Because it is not stabilized against the upper trunk or occiput, it does not provide any mechanical restriction of head motion. It can provide some warmth and comfort for patients with muscle strain. The collar is usually made from a block of foam rubber material that may be contoured around the chin. The foam is covered with a stockinette material, and Velcro is added to the ends to provide closure.

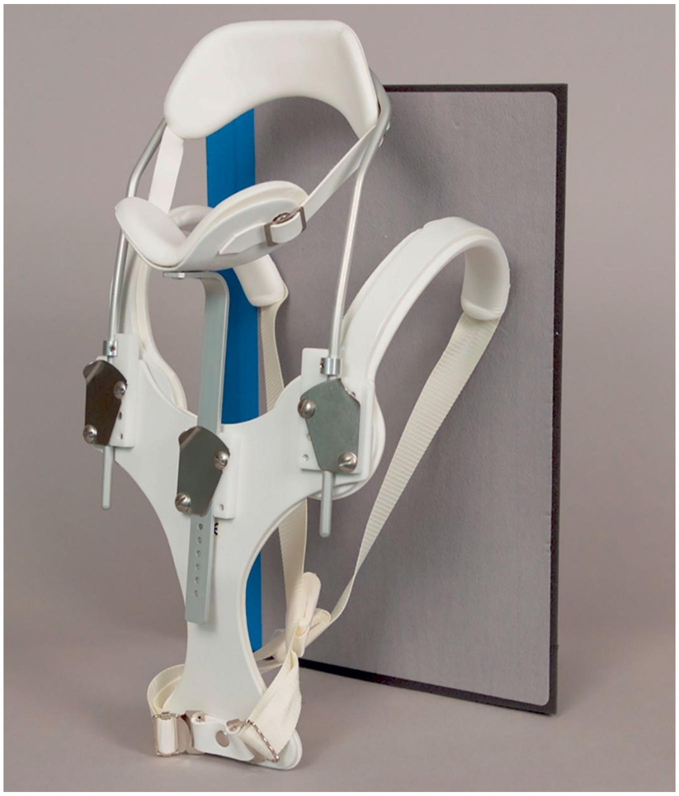
Cervicothoracic Orthoses

Type: Sternal Occipital Mandibular Immobilizer

Biomechanics. The sternal occipital mandibular immobilizer (SOMI) (Fig. 14.7) provides control of flexion, extension, lateral bending, and rotation of the cervical spine. Pressure systems are used for control of motion and to provide slight distraction for immobilization of the spine. A benefit of the SOMI orthosis is that it can be donned while the patient is in the supine position. The SOMI is a good choice for patients who are restricted to bed because there are no posterior rods to interfere with the comfort of the patient. A headband can be added so that the chin piece can be removed. This maintains stability but improves accessibility for daily hygiene and eating.

Design and Fabrication. The SOMI is prefabricated, consisting of a cervical portion with removable chin piece and bars that curve over the shoulders. Also used are posts that fixate the cervical portion to the sternal portion of the orthosis. The anterior section supports the mandible and rests on the superior edge of the sternum, with the inferior anterior edge terminating at the level of the xiphoid. The posterior aspect of the orthosis supports the head at the occipital level.

Indications. The SOMI is used primarily for cervical sprains, strains, or stable fractures with intact ligaments. It can also be used for protection and for limited mobility during the healing process in the postoperative patient. It can allow for control of cervical flexion in C1 to C3 levels, which is valuable with atlanto-axial instability and neural arch fractures.



• **Fig. 14.7** Sternal occipital mandibular immobilizer.

Contraindications. This orthosis is not indicated for unstable fractures with ligament instability.

Type: Four-Poster Orthosis

The four-poster type is a rigid cervical orthosis with anterior and posterior sections consisting of pads that lie on the chest and are connected by leather straps. The struts on the anterior and posterior sections are adjustable in height. Straps are used to connect the occipital and mandibular support pieces by way of the over-the-shoulder method.

Also note that some cervical orthoses (e.g., Aspen, Philadelphia, Miami J) can also incorporate a sternal extension addition, which converts them from a cervical orthosis to a cervicothoracic orthosis.

Type: Thermoplastic Minerva Body Jacket

Biomechanics. The Minerva orthosis provides flexion, extension, lateral bending, and rotational control of the cervical region. Pressure systems are used for control of motion and to provide slight distraction for immobilization of the cervical spine.

Design and Fabrication. The Minerva device was originally a plastic cast that fell out of favor with the introduction of the Halo vest. A thermoplastic Minerva body jacket was later introduced and considered easier to wear than the original design. A more recent design of the Minerva device is a prefabricated Minerva body jacket.[1]

The Minerva body jacket is a less invasive orthosis compared with the Halo vest because it does not utilize pins for immobilization. The orthosis encloses the posterior skull and consists of a band around the forehead and a mandibular pad. It extends down the neck and encompasses much of the thoracic region to the inferior costal margin.[9]

Indications. The Minerva orthosis is used for the management of cervical spine instability, although there is disagreement at which level of upper cervical immobilization it provides.

Contraindications and Complications. Limited motion can cause difficulty with mobility and self-care while using the Minerva orthosis. As with most spinal orthoses, prolonged and excessive use increases risk of development of pressure injuries, soft tissue breakdown, and spinal musculature weakness.[12] Because of its mandibular pad, the Minerva may also limit the motion of the hyoid bone, potentially creating difficulty with swallowing.[1] Some consider the Minerva orthosis to be as effective as the Halo vest for fractures in the upper cervical spine, while others recommend utilizing the Minerva in cases of fractures below C2. Although more consistent data are needed regarding the level of immobilization at the upper cervical spine, the Minerva brace remains an alternative to the more invasive Halo vest.[1]

Special Considerations. Because of increased comfort and reduced weight compared with the Halo vest, the Minerva orthosis may be preferred, particularly in younger children. However, the Halo vest is more commonly used for unstable fractures of the cervical spine for greater motion control.[9]

Type: Yale Cervicothoracic Orthosis

The Yale cervicothoracic orthosis is a foam Philadelphia collar that is reinforced with fiberglass extensions down the anterior and posterior thorax and includes straps that traverse under the axilla.[9,20] It also differs from the Philadelphia collar in that the occipital piece can extend higher up for further support.[9] It provides limitation of motion postsurgery in the lower and middle parts of the cervical spine.[21]

Cervicothoracolumbosacral Orthoses

Type: Milwaukee Orthosis

Biomechanics. The Milwaukee orthosis is used for scoliosis management (scoliosis is discussed later in this chapter). The Milwaukee provides control of flexion, extension, and lateral bending of the cervical, thoracic, and lumbar spine. It also provides some rotational control of the thoracic and lumbar spine. Pressure systems are used for control of motion and to provide correction for the spine. The Milwaukee is a good choice for patients who need correction in the higher thoracic region of the spine.

Design and Fabrication. The Milwaukee is custom made, consisting of a cervical portion with the option for a removable cervical ring. Also used is the thoracolumbar section of the orthosis, in which the correction of the lower thoracic and lumbar spines is achieved. Pads strapped to the bars apply transverse load to the ribs and spine to correct abnormal curvatures in persons with abnormal curvatures of the spine (e.g., scoliosis and thoracic Scheuermann disease kyphosis).[9]

Indications. The Milwaukee brace is used primarily for scoliotic management of the high thoracic curves along with thoracic and lumbar curves of the spine.

Contraindications. This orthosis is not indicated for lower thoracic and lumbar curves only. With lower thoracic and lumbar curves, a thoracolumbar orthosis could be used without the use of the cervical component.

• **Fig. 14.8** Thoracolumbosacral orthosis (prefabricated).

Thoracolumbosacral Orthoses

Type: Thoracolumbosacral Orthosis (Prefabricated)

Biomechanics. The TLSO (prefabricated) (Fig. 14.8) provides control of flexion, extension, lateral bending, and rotation using three-point pressure systems and circumferential compression.

Design and Fabrication. This orthosis can be designed in modular forms, with anterior and posterior sections connected by padded lateral panels and fastened with Velcro straps or pulley systems. Many of these are covered in breathable fabric and have a variety of different shapes and options, such as sternal pads or shoulder straps.

Indications. This orthosis can be used for treatment of traumatic or pathological spinal fractures in the middle to lower thoracic region or lumbar region.

Contraindications. These include a body habitus that is obese with a pendulous abdomen, excessive lordosis, or a need for increased lateral stability.

Special Considerations. Cost is reduced in some models because of the mass production of prefabricated modules. The use of breathable material can increase compliance with wearing.

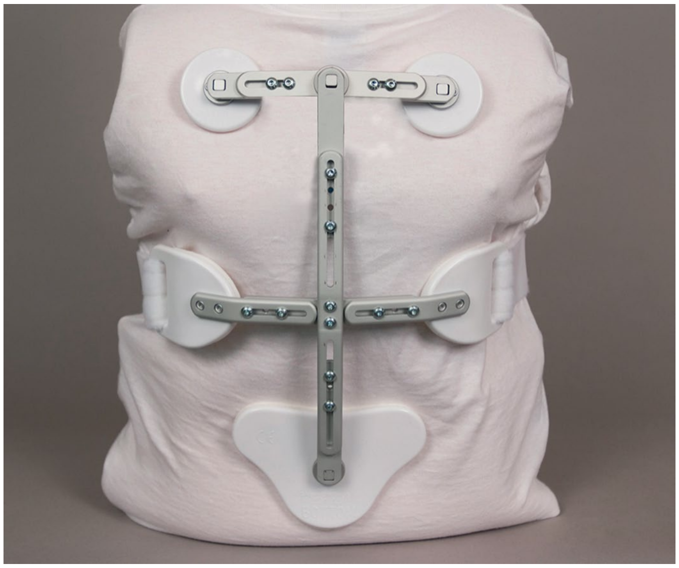
Type: Thoracolumbosacral Orthosis (Custom-Fabricated Body Jacket)

Biomechanics. The body jacket (Fig. 14.9) provides control of flexion, extension, lateral bending, and rotation. It uses three-point pressure systems and circumferential compression.

Design and Fabrication. It is molded to fit the patient and designed for patient needs. Anterior trim lines are usually located inferior to the sternal notch and superior to the pubic symphysis. The posterior trim lines have a superior border at the spine of the scapula and an inferior border at the level of the coccyx. These trim lines are adjusted during fitting to allow patients to sit comfortably and to use their arms as much as possible without compromising the function of the orthosis.

Indications. This orthosis can be used for treatment of traumatic or pathological spinal fractures in the middle to lower thoracic region or lumbar region. Most are used for postsurgical management of fractures, such as compression, chance, or burst. The brace is also used after surgical correction of spondylolisthesis, scoliosis, spinal stenosis, herniated discs, and disc infections.

• **Fig. 14.9** Thoracolumbosacral orthosis (custom-fabricated body jacket).



• **Fig. 14.10** Cruciform anterior spinal hyperextension thoracolumbosacral orthosis.

Contraindications. These include application of the orthosis over a chest tube, colostomy, or large dressings.

Special Considerations. Care must be taken to ensure that contact is maximized to decrease the pressure in any one area. Changes in the trim line must be made in small increments to prevent loss of control in terms of both leverage and tissue control. Ventilating holes are often made to improve airflow. Other factors to be considered when making the design include patients who might attempt to remove the orthosis when out of bed. The orthosis can be made with a posterior opening to reduce the risk.

Type: Cruciform Anterior Spinal Hyperextension Thoracolumbosacral Orthosis

Biomechanics. The CASH TLSO (Fig. 14.10) provides flexion control for the lower thoracic and lumbar regions. It accomplishes this by way of a three-point pressure system. The system consists of posteriorly directed forces through a sternal and suprapubic pad and an anteriorly directed force applied through a thoracolumbar pad attached to a strap that extends to the horizontal anterior bar.

• **Fig. 14.11** Jewett hyperextension thoracolumbosacral orthosis.

Design and Fabrication. The CASH is prefabricated, consisting of an anterior frame in the form of a cross, from which pads are attached laterally on a horizontal bar and at the sternal and suprapubic areas. A thoracolumbar pad is attached to a strap that extends to the lateral sections of the horizontal bar and adjusts the tension on the body. When properly fitted, the sternal pad is half an inch below the sternal notch, and the suprapubic pad is half an inch above the symphysis pubis.

Indications. This orthosis is used primarily for the treatment of mild compression fractures of the lower thoracic and thoracolumbar regions. It can also be used for kyphosis reduction and reduction in pain secondary to osteoporotic fractures.

Contraindications. The CASH is not indicated for unstable fractures or burst fractures. It is also contraindicated when extension is prohibited because the posterior elements may contribute to excessive hyperextension.

Special Considerations. Excessive pressure on the sternum can result in poor compliance with wearing schedule. Subclavicular pads may be added to help distribute this pressure.

Type: Jewett Hyperextension Thoracolumbosacral Orthosis

Biomechanics. The Jewett hyperextension TLSO (Fig. 14.11) provides flexion control for the lower thoracic and lumbar regions. This is done with a three-point pressure system consisting of posteriorly directed forces through a sternal and suprapubic pad and an anteriorly directed force applied through a thoracolumbar pad attached to a strap that extends to the lateral uprights.

Design and Fabrication. It is prefabricated, consisting of an anterior and lateral frame to which pads are attached laterally on and at the sternal and suprapubic areas. A thoracolumbar pad is attached to a strap that extends to the lateral uprights and adjusts the tension on the body. When properly fitted, the sternal pad is half an inch below the sternal notch, and the suprapubic pad is half an inch above the symphysis pubis.

Indications. The Jewett hyperextension TLSO is used primarily for the treatment of mild compression fractures of the lower thoracic and thoracolumbar regions. The Jewett has more lateral support than the CASH. It is also used in Scheurmann disease kyphosis and osteoporotic kyphosis.[9]

Contraindications. It is not indicated for unstable fractures or burst fractures. Caution should be taken with use for compression fracture in elderly patients with osteoporosis caused by excessive forces exerted on the lower lumbar spine. This is because there is a higher risk of posterior element fractures and exacerbation of degenerative arthritis.[9]

Special Considerations. Excessive pressure on the sternum might result in poor compliance with a wearing schedule. Subclavicular pads can be added to help distribute this pressure.

Type: Taylor and Knight-Taylor

Biomechanics. The Taylor and Knight-Taylor orthoses provide control of flexion, extension, and a minimal amount of axial rotation by means of the three-point pressure systems for each direction of motion. For example, flexion is controlled by the posteriorly directed forces applied through the axillary straps and the abdominal apron and an anteriorly directed force through the paraspinal uprights.

Design and Fabrication. The design of the Taylor orthosis consists of a posterior pelvic band extending past the midsagittal plane and across the sacral area. Two paraspinal uprights extend to the spine of the scapula. An apron front extends from the xiphoid to just above the pubic area. There are straps extending from the top of the posterior uprights around the posterior axilla to the scapular bar and forward to the apron. Other straps extend from the paraspinal uprights to the apron.

The Knight-Taylor orthosis has an additional thoracic band that extends from the uprights just below the inferior angle of the scapula to the midsagittal plane and a lateral upright on each side that connects the pelvic band and the thoracic band. These bands provide additional lateral support and motion control to the trunk.

Indications. These orthoses have been used for years for postsurgical support of traumatic fractures, spondylolisthesis, scoliosis, spinal stenosis, herniated discs, and disc infections. They can tighten the intraabdominal pressures and thus reduce lordosis by distending the lumbar spine. However, clinicians typically now prefer custom-molded TLSO body jackets because better control of position is obtained.

Contraindications. There are unstable fractures that require maximum stabilization.

Special Considerations. Pressure per square inch is higher over the thorax and abdomen because of the thinner width of the straps as compared with other orthoses, such as a TLSO body jacket.

Lumbosacral Orthoses

Type: Lumbosacral Corset

Biomechanics. The lumbosacral corset (Fig. 14.12) provides anterior and lateral trunk containment and assists in the elevation of intraabdominal pressure. Restriction of flexion and extension can be achieved with the addition of steel stays posteriorly.

• **Fig. 14.12** Lumbosacral corset.

Design and Fabrication. This orthosis is usually made from cloth that wraps around the torso and hips. Adjustments are done with laces on the sides, back, or front. Closure can be with hook and loop (Velcro) or hook and eye fasteners, or snaps. Many different styles are available in prefabricated sizes, usually in 2-inch increments, and are designed to fit the body circumference at the level of the hips. The orthosis can be adjusted for body type and proper fit by taking tucks in the cloth, as needed. Steel stays must be contoured to the body shape to encourage a reduction of lordosis or to accommodate a deformity. Custom corsets can be fabricated based on careful measurements of the individual patient.

Indications. This orthosis is the most frequently prescribed support for patients with low back pain.[34] It has been used for herniated disks and lumbar muscle strain and for the control of gross trunk motion for pain control after single-column compression fractures with one-third or less anterior height loss.[40]

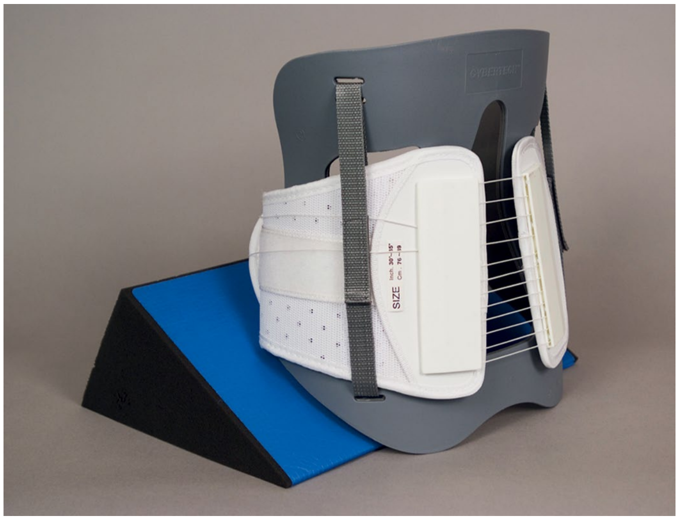
Contraindications. The orthosis should not be used for unstable fractures or for other fractures and conditions above the lower lumbar region.

Special Considerations. Long-term use of a lumbosacral corset can cause an increase in motion in the segments above or below the area controlled by the orthosis.[31] Muscle atrophy can also potentially occur after long-term use, causing an increased risk of reinjury. Patients can also have a psychological dependence on the support after injury.[25] There remains a lack of consensus regarding the efficacy of corsets for use in lower back pain.[9]

Type: Lumbosacral Chairback Orthoses

Biomechanics. This brace (Fig. 14.13) provides limitation of flexion, extension, and lateral flexion. It also provides elevation of intraabdominal pressure.

Design and Fabrication. This orthosis has a pelvic band that lies posteriorly and extends laterally to just anterior to the midsagittal line. Laterally, the ends fall midway between the iliac crest and the greater trochanter. The superior edge of the thoracic band is at the level of T9–T10 or just distal to the inferior angle of the scapulae.



• **Fig. 14.13** Lumbosacral orthosis chairback brace.

The pelvic and thoracic bands are connected by two paraspinal uprights posteriorly and a lateral upright on each side at the midsagittal line. Orthoses can be fabricated from a traditional aluminum frame covered in leather or thermoplastic material molded into the same shape. Straps are connected in a variety of ways to the frame, providing attachment to the anterior apron front.

Indications. This brace is often used for lower lumbar pathological conditions, including degenerative disc disease, herniated disc, spondylolisthesis, and mechanical low back pain, and for postsurgical supports for lumbar laminectomies, fusions, or discectomies.[34]

Contraindications. These are unstable fractures or conditions in the upper lumbar or thoracic area.

Special Considerations. Adequate clearance of the paraspinal uprights is required to allow for some reduction of lumbar lordosis when the anterior apron is tightened and while the patient is sitting. Clearance on the lateral uprights over the iliac crests is also an area to be monitored.

Sacroiliac Orthoses

Type: Sacroiliac Orthosis or Sacral Orthosis

Biomechanics. The sacroiliac orthosis provides anterior and lateral trunk containment and assists in the restriction of some pelvic flexion and extension. It also aids in compression of the pelvis.

Design and Fabrication. This orthosis is usually made from cloth that wraps around the pelvis and hips. Some models include laces on the side by which adjustments can be made, whereas others use straps for adjusting. Closure can be maintained with hook and loop (Velcro) fasteners, hook and eye fasteners, or snaps. Many different styles are available in prefabricated sizes, usually in 2-inch increments, and are designed to fit the circumference of the body at the level of the hips. The orthosis can be adjusted for body type and proper fit by taking tucks in the cloth, as needed. Custom orthoses can be fabricated based on careful measurements of the individual patient.

Indications. This orthosis is the most frequently prescribed as a support for patients with pelvic or pubic symphysis fractures/strains. It is useful to control motion and for pain control.

Contraindications. The orthosis should not be used for unstable fractures or for fractures or conditions in the lumbar region.

Scoliosis

Idiopathic (infantile, juvenile, adolescent), congenital, and neuromuscular scoliosis have different etiologies, treatment approaches, and outcomes. Idiopathic scoliosis is the most common form.[35] Idiopathic infantile scoliosis is typically described from birth to 3 years of age, juvenile is from 4 years until the onset of puberty, and the adolescent type is from puberty to closure of the facets. With idiopathic scoliosis, the evaluation should reveal no anomalous vertebrae, spinal tumors, or other neurological abnormalities. Most cases remain stable for a long period and progress late in life when osteoporosis and degenerative spinal conditions normally have their onset. Progressive curves need to be treated, but there is not adequate evidence that scoliosis can be treated by electrical stimulation, nutritional supplementation, exercise, or chiropractic treatment. There is strong evidence to indicate that an orthosis can slow the progression of idiopathic scoliosis, and therefore it is the nonoperative treatment of choice.[39]

Juvenile idiopathic scoliosis is more likely to be associated with adult cor pulmonale and death. Treatment should begin when curves reach approximately 25 degrees. Because thoracic curves predominate, the Milwaukee brace may be more effective than the TLSO. The Milwaukee (CTLSO) brace has a pelvic section with close contact with the iliac crest and lumbar spine. Three uprights (one anterior and two posterior) typically connect to a neck ring, throat mold, and occipital pad. The Boston brace is a TLSO that uses symmetrical standardized modules, eliminating the need for casting. It extends from below the breast to the beginning of the pelvic area and below the scapulae posteriorly. It maintains flexion of the lumbar area by increasing pressure on the abdomen and is a popular TLSO brace. Adolescent idiopathic scoliosis is the most common type for which an orthosis is indicated, usually for curves between 25 and 45 degrees. Curves with an apex at T9 or lower can be managed with a TLSO. Curves with a higher apex require a Milwaukee brace. Single lumbar curves are treated with a lumbosacral orthosis.[14] Congenital scoliosis is secondary to a vertebral anomaly that is present at birth. Failure of part of the vertebrae to form (e.g., hemivertebrae), failure of the vertebrae to properly segment (block vertebrae), or a combination of both can occur. Congenital scoliosis is associated with abnormal development in the embryo, and associated developmental abnormalities in other organ systems should be considered, especially in the renal, urinary, and cardiac systems.[29]

Neuromuscular diseases are also associated with scoliosis. The prevalence of scoliosis in this population is much higher than with idiopathic scoliosis, from 25% to 100%. In pediatric patients with a spinal cord injury, almost 100% develop scoliosis. In general, there is a significant chance of progression in the presence of severe neurological disease. In adults, however, scoliosis curvatures tend to be relatively benign and of the C-type appearance and are less likely to progress to the extent that they cause clinical cardiopulmonary problems. Progression of the curve can occur in adulthood, which is typical for scoliosis in general. Spasticity or flaccidity can be present, depending on whether there is upper or lower motor neuron involvement. Multisystem involvement is more common in this group because these diseases are not isolated to the spinal column. Consideration should also be made for the presence of contractures, hip dislocations, sensory abnormalities, mental impairment, and pressure injuries.[24]

Scoliosis can continue to progress despite the proper use of an orthosis, and in these cases appropriate surgical referrals should be made. An important factor to consider before surgery is the pulmonary function in a patient with neuromuscular disease. Before surgery is considered, the forced vital capacity and forced expiratory volume in 1 second should be at least 40% of that predicted for the patient's age. Fusions are delayed as long as possible in an attempt to achieve maximal spinal growth (>10 years of age). Declining pulmonary function, however, is a consideration for performing surgery earlier.[30] Duval-Beaupere[13] followed the long-term progression of idiopathic scoliosis and noted that curve progression accelerated during growth spurts. He noted that the younger the child, the higher the risk of curve progression because of a greater amount of growth that remained. It has also been shown that the greater the curve, the more likely the curve would increase.[23] Curves measured from 5 to 29 degrees and from 20 to 29 degrees progressed in almost 100% of the patients. Approximately 50% of the curves from 5 to 19 degrees appeared to progress. Curve progression has been explained with the Euler theory of elastic buckling of a slender column.[37] Axial compressive forces evidently cause a column to buckle. This is associated with height growth and weight gain, especially because of increased upper limb weight during growth spurts. An increase in height and weight commonly occurs together and might synergistically promote curve progression. It has been noted by us and others that the condition of a child with a large curve is more likely to progress than that of a child with a small curve.

The timing for surgery in a child with scoliosis is controversial. A child who has a curve greater than 45 degrees, a child who is still growing, or a child who cannot or does not wear a brace is at a greater risk of curve progression and may be considered for surgery.

Orthoses for Scoliosis (Boston Brace, Miami Orthosis, Wilmington Brace)

Type: Thoracolumbosacral Low-Profile Scoliosis Orthoses

Biomechanics. The TLSO low-profile scoliosis orthosis (Fig. 14.14) provides dynamic action using three principles (endpoint control, transverse loading, and curve correction) to prevent curve progression and to stabilize the spine.

Design and Fabrication. The use of orthoses or other devices to halt the curve progression of structural scoliosis has been reported as far back as Hippocrates.[27] Many different types of orthoses have been described in the literature. The one that stands out as being the most successful is the Milwaukee orthosis, which is described earlier in the chapter.

The effective nonoperative treatment of idiopathic scoliosis with a low-profile TLSO has been demonstrated over the past 30 years. The most common of these orthoses is the Boston brace, introduced by Hall and Miller[17] in 1975.[4] This system is available in prefabricated modules that come in 30 sizes and can be ordered by measurement; they are then custom-fit to the patient. Modules can be used to fit approximately 85% of patients. Six of these sizes will fit approximately 60% of patients requiring an orthosis.[16] The orthosis can also be custom fabricated from a mold of the patient's body. Trim lines are established on the basis of the patient's curve; they are designed to provide pressure in specified areas to maximize corrective forces and be less visible under the patient's clothing.

Indications. Both Boston and Charleston orthoses are indicated in patients with an immature skeleton and documented progression of a thoracic or thoracolumbar idiopathic scoliosis that measures 25 to 35 degrees (measured by the Cobb method) and has an apex of T7 or lower.[7] A study in 1996 demonstrated that Boston braces are more effective in treating curves over 36 degrees and can address multiple curves in comparison to the Charleston orthoses. Charleston braces are usually worn at night-time to assist with straightening scoliotic curves when supine.

• **Fig. 14.14** Thoracolumbosacral low-profile scoliosis orthosis.

Contraindications. The orthosis is contraindicated in patients who have curves that measure more than 40 degrees and who are skeletally immature or who have curves greater than 50 degrees after the end of growth. Both of these types of patients are typically candidates for surgery.[8]

Special Considerations. Idiopathic scoliosis occurs mostly in adolescent females. Treatment regimens, both nonoperative and operative, are major events for patients and their families. To be successful, the treatment team must be sensitive, supportive, and honest, communicating accurate information to patients and their families. The effectiveness of any orthotic system depends on compliance with the wearing schedule. Most patients should wear the orthosis 23 to 24 hours per day for it to be effective. Some physicians allow time out of the orthosis to participate in athletic activities or swimming and some special occasions, and this seems to improve acceptance and compliance.

Emerging Technology

Computer-Aided Design and Computer-Aided Manufacturing

Technology is available to help the practitioner improve efficiency in design and fabrication and to reduce the invasiveness of orthotic measurement of patients. The development of computer-aided design (CAD) and computer-aided manufacturing (CAM) has allowed the fabrication of orthoses today in less time

• **Fig. 14.15** BioScanner handheld laser scanner system.

than it took only a few years ago. The BioScanner BioSculptor (www.biosculptor.com) is one such CAD-CAM system (Fig. 14.15). It combines CAD, laser scanning, 3D imagery, and motion-tracking technology to design orthotic and prosthetic devices. This is an alternative to traditional casting methods. The body part involved is scanned with a handheld scanning wand, which uses a motion-tracking device embedded in the scanner. The wand is passed over the body part. A 3D image of the body part is transmitted to the computer, and the software interprets the data. A miniature transmitter is placed on the body to accommodate for any movement. The BioScanner is able to image negative and positive models, allowing the clinician to use the clinical techniques required for each patient. With scan-through-glass technology that the company provides for use with its scanner, the clinician can position the body horizontally for a TLSO. The BioScanner automatically corrects the refraction. The precision of the BioScanner captures shapes to an accuracy of 0.178 mm. It provides the clinician with detailed and accurate 3D measurements. Patients can be scanned for spinal jackets without movement from supine to prone positions. An option in the computer software allows one hemisphere to be scanned, and then the computer develops the other hemisphere to form a complete image. The older plaster cast methods are becoming obsolete because of the following advantages of the scanning system: It is not messy, can be done quickly, minimizes pain for the patient, and is more accurate. Modifications to these scanned computerized models can then be made before fabrication of the device begins. Accuracy of measurement gives detailed surface information often lost with a cast or mechanical digitizer. The patient's scan is maintained in the computer database, allowing for rapid refitting and medical justification of new devices by showing volumetric changes.

• **Fig. 14.16** CMF SpinaLogic bone stimulator. (Courtesy DJO Global Inc, Vista, CA.)

Bone Stimulation

Portable, battery-powered, microcontrolled, noninvasive bone growth stimulators are available for adjunct electromagnetic treatment to primary lumbar spinal fusion surgery for one or two levels (Fig. 14.16). There are different bone stimulators on the market in use today. Some are used in conjunction with spinal orthoses and do not interfere with the control that the orthosis provides nor with the treatment protocol set by the physician. Some of the benefits of these bone stimulators are as follows: They are lightweight, comfortable, and easy to use; they can be used after an anterior or posterior approach in surgery; and they are noninvasive.

Three-Dimensional Clinical Ultrasound

Recent advances in 3D clinical ultrasound have allowed estimation of the spinous process angle (SPA) in patients with adolescent idiopathic scoliosis. This in turn has been able to provide orthotists a fast and safe method to assess the SPA in real time and determine the optimal placement of pressure pads to maximize the effectiveness of the orthotic in correcting the spinal deformity.[22]

Summary

The proper prescription, construction, and fitting of a spinal orthosis are complicated processes. For traumatic injuries, Table 14.2 provides a helpful guide for spinal orthotic prescription. A complete, clear, and agreed upon plan of care should be formulated. Success is likely to be limited if there is no agreement or understanding of the process and goals. Experienced and knowledgeable providers working in a team approach provide the maximum likelihood that the orthosis will contribute to the overall therapeutic goals of the patient.

TABLE 14.2 Orthoses for Spinal Trauma and Postoperative Care

Level	Type	Mechanism of Injury	Clinical Significance	Orthosis
C1	Jefferson	Axial load	Triplanar instability	Halo vest
C2	Hangman	Hyperextension plus distraction	Traumatic spondylolisthesis, triplanar instability	Halo vest
C2	Odontoid	Shear plus compression	Type 1: stable Types II and III: unstable	Halo vest
C3–C7	Anterior compression	Hyperflexion	C5 most common	Rigid collar
C3–C7	Whiplash	Hyperextension	Soft tissue injury, commonly to anterior longitudinal ligament Long-term risk for cervical flexion	Soft collar
T1–T8	Denis I: Anterior compression	Flexion plus compression	Most injuries at T12–L2 levels Damage limited to anterior column in most cases. Posterior ligamentous injury will likely result in instability.	Corsets
T1–T8	Denis II: Burst	Compression plus flexion	Superior end-plate damage is more common than anterior/middle column damage. Retropulsion of fragments from posterior wall may occur.	Jewett braces for milder injuries TLSO for moderate-severe injury
T9–T12	Denis III: chance and slice	Flexion plus distraction "seat belt injury"	Chance: Middle and posterior column damage to vertebral body Slice: Middle and posterior column damage to intervertebral disc; surgery indicated	TLSO braces
L1–L2	Denis IV: fracture dislocation	Translation, flexion, rotation, shear	Complete disruption of anterior, middle, and posterior columns often occurs; surgery indicated	Braces usually not indicated postop
L3–L5	Spondylolysis/ Spondylolisthesis	Sports related in athletes. "Wear and tear" in adults. Cause of chronic low blood pressure	L4–L5/L5–S1 injury most common Spondylolisthesis requires posterior pelvic tilt in orthosis	Custom LSO or TLSO

LSO, Lumbrosacral orthosis; *TLSO*, thoracolumbosacral orthosis.
From Malas BS, Meade KP, Patwardhan AG, et al. Orthoses for spinal trauma and postoperative care. In: Hsu JD, Michael JW, Fisk JR, eds. *Atlas of Orthotics and Assistive Devices.* 4th ed. Mosby; 2008.

Key References

1. Agabegi SS, Asghar FA, Herkowitz HN. Spinal orthoses. *J Am Acad Orthop Surg*. 2010;18(11):657-667.
4. Boston Brace International. [Website]. Accessed December 2018. www.bostonbrace.com.
8. Cassella MC, Hall JE. Current treatment approaches in the nonoperative and operative management of adolescent idiopathic scoliosis. *Phys Ther*. 1991;71:897-909.
11. Denis F. Spinal instability as defined by the three-column spine concept in acute spinal trauma. *Clin Orthop*. 1984;189:65-76.
12. Dinescu LI, Crimmins M, Shmargun M, et al. *Cervical, Thoracic, and Lumbosacral Orthoses*. PM&R Knowledge Now. 2021. Accessed July 16, 2024. https://now.aapmr.org/cervical-thoracic-and-lumbosacral-orthoses/.
14. Fisher SV, Winter RB. Spinal orthosis in rehabilitation. In Braddom RL, ed. *Physical Medicine and Rehabilitation*. Saunders; 1996.
15. Gavin TM, Carandang G, Havey R, et al. Biomechanical analysis of cervical orthoses in flexion and extension: a comparison of cervical collars and cervical thoracic orthoses. *J Rehabil Res Dev*. 2003;40:527-537.
17. Hall JE, Miller W, Shuman W, et al. A refined concept in the orthotic management of idiopathic scoliosis. *Prosthet Orthot Int*. 1975;29:7-13.
19. Horodyski M, DiPaola CP, Conrad BP, et al. Cervical collars are insufficient for immobilizing an unstable cervical spine injury. *J Emerg Med*. 2011;41:513-519.
21. Karimi MT, Kamali M, Fatoye F. Evaluation of the efficiency of cervical orthoses on cervical fracture: a review of literature. *J Craniovertebr Junction Spine*. 2016;7(1):13-19.

24. Lonstein JE, Carlson M. The prediction of curve progression in untreated idiopathic scoliosis during growth. *J Bone Joint Surg Am*. 1984;66:1061-1071.
25. Luskin R, Berger N. Atlas of Orthotics: *Biomechanical Principles and Application*. American Academy of Orthopaedic Surgeons. Mosby; 1975.
28. Malas BS, Meade KP, Patwardhan AG, et al. Orthoses for spinal trauma and postoperative care. In: Hsu JD, Michael JW, Fisk JR, eds. *Atlas of Orthotics and Assistive Devices*. 4th ed. Mosby; 2008.
30. Monstein JK. Orthosis for spinal deformities. In: Goldberg B, Hsu JD, eds. *Atlas of Orthoses and Assistive Devices*. 3rd ed. Mosby; 1997.
33. Panjabi MM, Takata K, Goel V, et al. Thoracic human vertebrae, quantitative three-dimensional anatomy. *Spine*. 1991;16:888-901.
34. Perry J. The use of external support in the treatment of low back pain. *J Bone Joint Surg Am*. 1970;52:1440-1442.
35. Tachdejian MO. *Pediatric Orthopedics*. 2nd ed. Saunders; 1990.
36. Taitsman LA, Altman DT, Hecht AC, et al. Complications of cervical halo-vest orthoses in elderly patients. *Orthopedics*. 2008;31:446.
39. Weinstein SL, Dolan LA, Wright JC, et al. Effects of bracing in adolescents with idiopathic scoliosis. *N Engl J Med*. 2013;369:1512-1521.
40. White AA, Panjabi MM. Extension injuries. In: White AA, Panjabi MM, eds. *Clinical Biomechanics of the Spine*. 2nd ed. Lippincott; 1990.

Visit Elsevier eBooks+ at eBooks.Health.Elsevier.com for a complete list of references.

References

1. Agabegi SS, Asghar FA, Herkowitz HN. Spinal orthoses. *J Am Acad Orthop Surg*. 2010;18(11):657-667.
2. Andry N. *Orthopaedia* [facsimile reproduction of the first edition in English, London, 1743]. Lippincott; 1961.
3. Anonymous. *Dorland's Medical Dictionary*. 24th ed. Saunders; 1989.
4. Boston Brace International. [Website]. Accessed December 2018. www.bostonbrace.com.
5. Bransford RJ, Stevens DW, Uveil S, et al. Halo vest treatment of cervical spine injuries: a success and survivorship analysis. *Spine*. 2009;34(15):1561-1566.
6. Bunch WH, Keagy R. *Principles of Orthotic Treatment*. Mosby; 1975.
7. Carr WA, Moe JH, Winter RB, et al. Treatment of idiopathic scoliosis in the Milwaukee brace: long term results. *J Bone Joint Surg Am*. 1980;62:599-612.
8. Cassella MC, Hall JE. Current treatment approaches in the nonoperative and operative management of adolescent idiopathic scoliosis. *Phys Ther*. 1991;71:897-909.
9. Cuccurullo SJ. *Physical Medicine and Board Review*. 4th ed. Demos Medical; 2020:527-529, 531-532.
10. Denis F. The three column spine and its significance in the classification of acute thoracolumbar spinal injuries. *Spine*. 1983;8:817-831.
11. Denis F. Spinal instability as defined by the three-column spine concept in acute spinal trauma. *Clin Orthop*. 1984;189:65-76.
12. Dinescu LI, Crimmins M, Shmargun M, et al. *Cervical, Thoracic, and Lumbosacral Orthoses*. PM&R Knowledge Now. 2021. Accessed July 16, 2024. https://now.aapmr.org/cervical-thoracic-and-lumbosacral-orthoses/.
13. Duval-Beaupere G. Pathogenic relationship between scoliosis and growth. In: Zorab BA, ed. *Proceedings of the Third Symposiums on Scoliosis and Growth*. Churchill Livingstone; 1971.
14. Fisher SV, Winter RB. Spinal orthosis in rehabilitation. In Braddom RL, ed. *Physical Medicine and Rehabilitation*. Saunders; 1996.
15. Gavin TM, Carandang G, Havey R, et al. Biomechanical analysis of cervical orthoses in flexion and extension: a comparison of cervical collars and cervical thoracic orthoses. *J Rehabil Res Dev*. 2003;40:527-537.
16. Hall JE, Miller ME, Cassella MC, et al. *Manual for the Boston Brace Workshop*. Children's Hospital; 1976.
17. Hall JE, Miller W, Shuman W, et al. A refined concept in the orthotic management of idiopathic scoliosis. *Prosthet Orthot Int*. 1975;29:7-13.
18. Hall-Craggs EC. The back and spinal cord. In: Tyler NC, ed. *Anatomy as a Basis for Clinical Medicine*. Urban & Schwarzenberg; 1985.
19. Horodyski M, DiPaola CP, Conrad BP, et al. Cervical collars are insufficient for immobilizing an unstable cervical spine injury. *J Emerg Med*. 2011;41:513-519.
20. Johnson RM, Hart DL, Owen JR, et al. The Yale cervical orthosis: an evaluation of its effectiveness in restricting cervical motion in normal subjects and a comparison with other cervical orthoses. *Phys Ther*. 1978;58(7):865-871.
21. Karimi MT, Kamali M, Fatoye F. Evaluation of the efficiency of cervical orthoses on cervical fracture: a review of literature. *J Craniovertebr Junction Spine*. 2016;7(1):13-19.
22. Katz DE, Richards BS, Browne RH, et al. A comparison between the Boston brace and the Charleston bending brace in adolescent idiopathic scoliosis. *Spine*. 1997;22:1302-1312.
23. Lonstein JE. Orthoses for spinal deformities. In: Goldberg B, Hsu JD, eds. *Atlas of Orthotics: Biomechanical Principles and Application*. 2nd ed. Mosby-Year Book; 1985.
24. Lonstein JE, Carlson M. The prediction of curve progression in untreated idiopathic scoliosis during growth. *J Bone Joint Surg Am*. 1984;66:1061-1071.
25. Luskin R, Berger N. American Academy of Orthopaedic Surgeons: *Atlas of Orthotics: Biomechanical Principles and Application.* Mosby; 1975.
26. Maejercik S, Tashjian RZ, Biffl WL, et al. Halo vest immobilization in the elderly: a death sentence? *J Trauma*. 2005;59:350-358.
27. Maiman D, Millington P, Novak S, et al. The effects of the thermoplastic Minerva body jacket on cervical spine motion. *Neurosurgery*. 1989;25:363-368.
28. Malas BS, Meade KP, Patwardhan AG, et al. Orthoses for spinal trauma and postoperative care. In: Hsu JD, Michael JW, Fisk JR, eds. *Atlas of Orthotics and Assistive Devices*. 4th ed. Mosby; 2008.
29. Moe JH, Bradford DS, Winter RB, et al. *Scoliosis and Other Spinal Deformities*. Saunders; 1978.
30. Monstein JK. Orthosis for spinal deformities. In: Goldberg B, Hsu JD, eds. *Atlas of Orthoses and Assistive Devices*. 3rd ed. Mosby; 1997.
31. Norton P, Brown T. The immobilization efficiency of back braces: the effect on the posture and motion of the lumbosacral spine. *J Bone Joint Surg Am*. 1957;39:111-130.
32. Panagiotacopulos ND, Pope MH, Bloch R, et al. Water content in human intervertebral discs: part II. Viscoelastic behavior. *Spine*. 1987;12:918-924.
33. Panjabi MM, Takata K, Goel V, et al. Thoracic human vertebrae, quantitative three-dimensional anatomy. *Spine*. 1991;16:888-901.
34. Perry J. The use of external support in the treatment of low back pain. *J Bone Joint Surg Am*. 1970;52:1440-1442.
35. Tachdejian MO. *Pediatric Orthopedics*. 2nd ed. Saunders; 1990.
36. Taitsman LA, Altman DT, Hecht AC, et al. Complications of cervical halo-vest orthoses in elderly patients. *Orthopedics*. 2008;31:446.
37. Timoshenko S, Gere J. *Theory of Elastic Stability*. 2nd ed. McGraw-Hill; 1961.
38. Wambolt A, Spencer DL. A segmental analysis of the distribution of lumbar lordosis in the normal spine. *Orthop Trans*. 1987;11:92-93.
39. Weinstein SL, Dolan LA, Wright JC, et al. Effects of bracing in adolescents with idiopathic scoliosis. *N Engl J Med*. 2013;369:1512-1521.
40. White AA, Panjabi MM. Extension injuries. In: White AA, Panjabi MM, eds. *Clinical Biomechanics of the Spine*. 2nd ed. Lippincott; 1990.
41. White AA, Panjabi MM. The lumbar spine. In: White AA, Panjabi MM, eds. *Clinical Biomechanics of the Spine*. 2nd ed. Lippincott; 1990.